

Dual-Track UX: A User Experience Evaluation Method for B2B SaaS Development

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Abstract

User experience (UX) evaluation is critical for identifying user requirements and mitigating the risk of product failure during the development of interactive systems. This paper presents Dual-Track UX, a specialized UX evaluation method designed to address three key challenges in evaluating the UX of Business-to-Business (B2B) Software-as-a-Service (SaaS) products: (a) diversified user goals and operational patterns, (b) limited access to external users, and (c) the necessity for long-term evaluation. Dual-Track UX integrates heuristic evaluation, Goals-Signals-Metrics process, exploratory testing, and service design tools into a cohesive dual-track workflow. Track 1 identifies and prioritizes usability issues in the user interface, while Track 2 constructs UX metrics and monitors the effectiveness of user goal achievement over time. Preliminary validation through expert interviews highlighted the advantages, limitations, challenges in adoption, and opportunities for further refinement, paving the way for its broader application in B2B SaaS development.

CCS Concepts

• Human-centered computing → HCI design and evaluation methods.

Keywords

User Experience Evaluation Method, B2B, SaaS, Usability Heuristics, Service Design

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1 INTRODUCTION

Human-Computer Interaction (HCI) primarily focuses on designing and evaluating interactive systems intended for human use and exploring related phenomena [4]. The evaluation of UX is a core concern in the HCI domain as it offers valuable insights into

what makes interactive systems usable and capable of delivering a positive overall experience [18]. Especially during the development phase of interactive systems, UX evaluation is critical for identifying user requirements and mitigating the risk of product failure [23]. This paper focuses on the specific context of Business-to-Business (B2B) Software-as-a-Service (SaaS), an emerging software delivery model designed and iterated to meet the unique demands of business operations while offering seamless access to applications via any internet-connected device [22].

However, the characteristics of B2B SaaS product development complicate the UX evaluation in three key aspects:

(a) **Diversified user goals and operation patterns:** The diversity of stakeholders with different competencies, responsibilities, and workflows in business operations makes it more challenging to define user goals and operation patterns compared to Business-to-Consumer (B2C) enterprises [2, 7].

(b) **Limited external user access:** Concerns about confidentiality and recruitment ease often led companies to prefer internal over external user testing; thereby, indirect access to end-users may be prone to bias [12, 21].

(c) **Necessity of long-term evaluation:** The iterative nature of the SaaS delivery model also diminishes the effectiveness of single-time user testing, necessitating long-term user experience monitoring [13].

Our research on previous user experience evaluation methods (UXEMs) for B2B SaaS products revealed a lack of approaches that fully satisfy the development needs mentioned above. In contrast, some single UXEMs do meet these needs. First, while usability heuristic evaluation offers a cost-effective alternative to address the limited access to external users, monitoring long-term UX changes remains challenging [11]. Furthermore, the integrated approach of the HEART framework and Goals-Signals-Metrics (GSM) Process proposed by Google, developed initially for B2C E-commerce websites, is effective for long-term evaluation by supporting mapping user and business goals to specific UX metrics [19]. However, there remains a need for further research on how to identify goals, extract signals, and construct UX metrics in a context-specific way. In this regard, developing a new UXEM incorporating human-centered design tools to provide insights from the users' perspective becomes a potential solution.

Thus, the authors propose Dual-Track UX: a UXEM designed for the key challenges of UX evaluation for B2B SaaS products. Dual-Track UX integrates existing UXEMs with service design tools, organizing the evaluation process into two evaluation tracks: (1) heuristics evaluation for identifying usability issues of the user

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interface and (2) UX metrics for monitoring the extent to which user goals are achieved.

Interviews with UX experts in academia and industry were conducted to obtain feedback on the first version of Dual-Track UX. The main objective was to validate its usefulness and identify its advantages and disadvantages compared to existing UXEMs. In addition, it is necessary to understand the challenges of adopting Dual-Track UX in work environments so that users' perspectives can be fully considered in subsequent iterations. Lastly, specific suggestions for refinement were also obtained to be reflected in the next version.

The contributions of this research to the human-computer interaction domain are mainly three:

(1) The authors integrate existing theories and methods to provide a comprehensive UXEM for B2B SaaS product development. They address the three significant challenges when evaluating the UX of B2B SaaS products.

(2) The authors preliminarily validated Dual-Track UX by conducting interviews with 3 UX experts in academia and industry and subsequent thematic analysis. The results reveal the advantages and disadvantages, challenges in adopting this method, and further improvement opportunities.

(3) This study also consolidates a practical approach to UXEM development, which may serve as a reference for other researchers developing UXEMs in related domains.

This paper is structured as follows: Section 2 introduces the theoretical background; Section 3 presents how Dual-Track UX is constructed; Section 4 illustrates its usage framework; Section 5 details the key results derived from the thematic analysis; Section 6 discusses contributions and limitations; Section 7 summarizes this study and outlines future work.

2 THEORETICAL BACKGROUND

This section explains the concepts of user experience, usability, and common methods and tools for user experience evaluation. Moreover, it presents works related to B2B SaaS UX evaluation and its correlation with them.

2.1 Definition of User Experience and Usability

According to ISO 9241-110:2010 (clause 2.15), user experience is defined as a person's perceptions and responses resulting from the use and/or anticipated use of a product, system, or service [8]. In comparison, usability, as defined by ISO 9241-11:2018, refers to the extent to which a system, product, or service can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction within a specified context of use [9]. While there is some ambiguity in ISO definitions regarding whether usability should be considered a component of user experience [1], this research tends to adopt the standpoint that usability is a subset of user experience, enabling UX evaluation to encompass both an objective product attribute perspective as well as a subjective user perception perspective.

2.2 Methods and Tools for UX Evaluation

Heuristics evaluation is a usability evaluation method that allows experts to review and provide feedback on user interfaces based

on a set of heuristics. It offers early and rapid feedback before conducting full-scale usability testing, which is an advantage over traditional usability tests [10].

In addition, when evaluating, the metric is recognized as the value that indicates a particular phenomenon, such as temperature, weight, height, time, and many others [20]. In the UX domain, a metric is used as an indicator that measures the user experience when interacting with a product, service, or system. With UX metrics, practitioners can obtain substantial and visualized user experience data and gain insights into areas needing improvement [24].

The service design tool "persona" is widely used to support UX evaluation as it identifies the goals of typical users and impacts UX by stimulating empathy and improving communication within the field of UX design teams [20].

User journey maps facilitate UX evaluation by providing a step-by-step visual representation of how users interact with a service over time, highlighting areas that may require attention or improvement in the follow-up version iteration [5].

2.3 Related Work

The Goals-Signals-Metrics (GSM) process developed by Google helps product teams construct UX metrics based on specific feature goals, enabling strategic focus, valuable insights, and a more predictable ROI (Return on Investment) [19]. The HEART Framework is recommended to be used in parallel to build metrics based on five dimensions of goals: Happiness (satisfaction or pleasure derived from using the product), Engagement (frequency, intensity, or depth of user interaction with the product), Adoption (new users who start using the product), Retention (the rate at which existing users continue using the product) and Task Success (efficiency and effectiveness of users accomplishing their tasks). However, further research is needed on how to identify goals, extract signals, and construct UX metrics in a context-specific way.

While FRAMUX-EV framework provides numerous UXEMs for agile development, it lacks effective integration to address the unique challenges faced by B2B SaaS [14]. By integrating UX into the iterative processes of agile projects, FRAMUX-EV addresses a critical gap in existing approaches, which often fail to account for UX artifacts and UX events adequately. Although the framework fully considers the need for long-term UX evaluation and user-centered workflows in iterative development and also incorporates expert review-based UX evaluation methods (such as Heuristic Evaluation) to address the issue of limited external user access effectively, there remains potential for further research on how to integrate user-centered workflows with UXEMs that do not involve full-process user engagement.

3 DUAL-TRACK UX: FIRST VERSION

As illustrated in Figure 1, the authors developed the first version of Dual-Track UX to address three key challenges of UX evaluation in B2B SaaS product development by incorporating existing methods and tools. To address (a) Diversified user goals and operation patterns, service design tools such as personas and user journey maps are employed to capture and visualize complex user goals and workflows. For (b) Limited external user access, heuristic evaluation and exploratory testing offer practical approaches to

evaluating UX with internal resources. Finally, (c) The Necessity of long-term UX evaluation is tackled through the GSM process, enabling practitioners to construct metrics derived from specific user goals and monitor UX change over time. These methods and tools are integrated into a dual-track workflow, as detailed below.

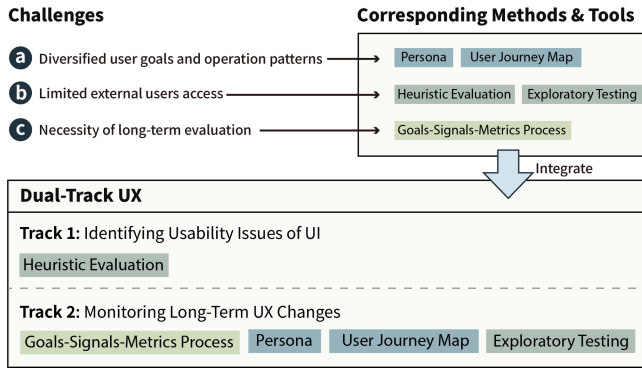


Figure 1: Integration of methods and tools

Track 1 adopts heuristic evaluation to help identify and prioritize usability issues of the user interface. Given the expertise required for heuristic evaluation, an enhanced workflow was developed to enable evaluators with limited experience to conduct the process effectively. The construction process involves four steps. First, the Extensional Classification approach [6] is applied to consolidate usability constructs from usability models by Eason, Shackel, Nielsen, ISO 9241-11, and ISO 9126 [15]. Second, to ensure broad applicability, key usability attributes are interpreted from two famous usability heuristics: Jakob Nielsen’s 10 Usability Heuristics [17] and the Principle of Consistency and Standards in User Interface Design [25]. Third, these attributes are mapped to their corresponding usability constructs for model alignment. Finally, a scale and coordinate system (Figure 2) adapted from Nielsen [16] is added to facilitate identifying and prioritizing usability issues based on severity and frequency.

Track 2 integrates the Goals-Signals-Metrics (GSM) process with service design tools and exploratory testing to construct UX metrics and monitor how effectively users achieve their goals over time. The GSM process, introduced as a foundation for evaluating long-term UX changes, aligns well with the iterative development cycles characteristic of the SaaS model. By incorporating service design tools—specifically personas and user journey maps—into the GSM process, this approach helps identify user goals and map them to corresponding signals observed throughout the user journey. Furthermore, exploratory testing, a widely used UXEM in internal settings, is integrated into this workflow to gather actionable user data.

4 USAGE FRAMEWORK OF DUAL-TRACK UX

This section provides a detailed guide on the practical usage framework of Dual-Track UX.

4.1 Track 1: Identifying Usability Issues of UI

For the practical application of Track 1, a minimum of three evaluators are required to individually assess each evaluation item in the “Rating” column of Table 1 after interacting with a B2B SaaS product. The ratings are assigned on a scale where 0 indicates “not a usability issue,” 1 denotes a “cosmetic issue,” 2 signifies a “minor issue,” 3 represents a “major issue,” and 4 indicates a “catastrophic issue.” Once individual assessments are completed, evaluators should aggregate their results and plot every code of evaluation item (e.g., LE1) within the coordinate system shown in Figure 2 based on the severity of each issue and the frequency with which it is encountered. In this system, problems closer to the upper-right corner represent higher-priority issues requiring immediate resolution. In contrast, those nearer to the lower-left corner are lower-priority and can be addressed later.

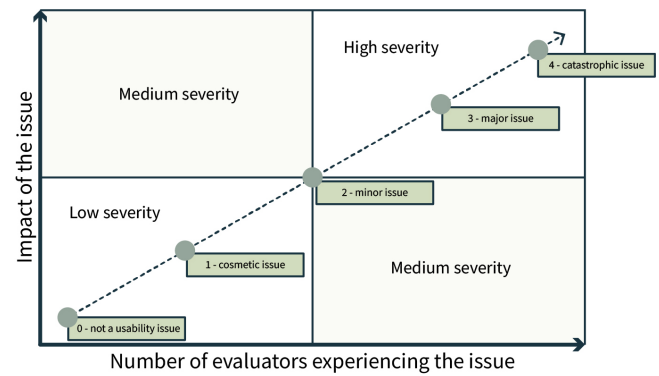


Figure 2: Coordinate system for prioritizing the severity of usability problems (Adapted from Nielsen [16])

4.2 Track 2: Monitoring Long-term UX Changes

As illustrated in Figure 3, when implementing Track 2, evaluators use personas to identify users’ goals by analyzing survey data, conducting user interviews, and examining preprocessed or manipulated web analytics data. Next, they create user journey maps to visualize users’ interaction with the system throughout the flow of a B2B SaaS service. Based on these maps, evaluators then identify signals by determining which phenomena reflect the achievement of user goals. Finally, they can construct user-centered UX metrics derived from these signals. Using these metrics, UX teams can monitor long-term UX changes by leveraging data collected from exploratory testing conducted by internal users.

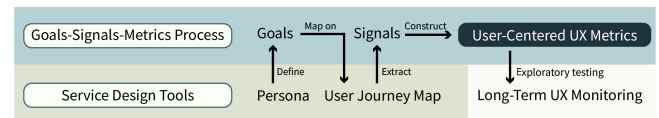


Figure 3: The Usage Framework of Track 2

Table 1: Usability Heuristic Scale (Created by the authors)

Usability Constructs	Description	Code	Attributes	Rating
Learnability	The capability of the software product to enable the user to learn its application, including first-time use and after long intervals of non-use.	LE	1. Language consistency 2. Language comprehensibility 3. UI Elements consistency 4. Visibility of system status 5. Minimization of memory load 6. Immediate feedback 7. Ease of search documentation 8. Contextual documentation 9. Concrete steps of documentation	
Understandability	The capability of the software product to enable the user to understand its suitability and how it can be used for tasks and conditions of use.	UN	1. User expectation 2. UI-Task consistency 3. Content and functional priorities 4. Information redundancy	
Operability	The capability of the software product to enable the user to operate and control it.	OP	1. Clear exit 2. Support of undo 3. Support of redo	
Effectiveness	The performance measure of a system to complete a specified task or goal successfully within time.	EE	1. Given constraints 2. Good defaults 3. Error message delivery 4. Confirmation before action 5. Authority to undo 6. Error message comprehensibility 7. Solution offerings	
Efficiency	The successful completion of a task by a system, relating to accuracy and completeness of the specified goal.	EI	1. Keyboard shortcuts 2. Touch gestures 3. Personalization 4. Customization	
Usability Compliance	The capability of the software product to adhere to standards, conventions, style guides, or regulations related to usability.	UC	1. Layout conventions 2. Consistency and standard of visual elements (font, background, color, UI, branding)	

5 VALIDITY ASSESSMENT

To validate this UXEM, the authors invited interviewees with at least 5 years of experience in UX design or research from academia and industry to participate in semi-structured interviews. Due to human networking and scheduling constraints, the authors could only conduct a validity assessment with three interviewees. Interviewee A from a small-and-middle-sized enterprise and Interviewee B from a large company are both UX managers with five years of experience developing B2B SaaS products in the industry. Interviewee C is currently an associate professor at a university with six years of industry experience and nine years of education and research in the field of UX. Subsection 5.1 introduces the interview procedure, and 5.2–5.5 provides a detailed presentation of the thematic analysis results and key findings.

5.1 Semi-structured Interview Procedure

Before starting, the authors sent a pdf file of the guide of this UXEM to the interviewees to familiarize them in advance with the scope and conceptual framework of this UXEM, as well as the specific steps in implementation. After this, semi-structured interviews are conducted using the online meeting software Zoom, following the procedure outlined in Table 2. The process begins with the interviewer introducing, explaining the purpose of the interview, and obtaining consent to record the session. Next, the interviewees are asked to provide background information about their experience in UX, as well as their preliminary understanding of UXEMs.

Following this, the interviewer spends approximately 30 minutes introducing Dual-Track UX's construction process and usage framework, regularly checking in with the interviewees to ensure their understanding. This introduction to the method is followed by questions regarding its advantages, disadvantages, challenges in adoption, and suggestions for improvements.

5.2 Thematic Analysis Results

The thematic analysis adopts Braun and Clarke's methodology [3] to identify the advantages, disadvantages, challenges in adoption, and potential improvements. As a result, the authors developed 26 codes and five overarching themes (Table 3).

5.3 Advantages and Disadvantages

One notable advantage of this UXEM is its ability to provide quantitative criteria that encompass both objective user interface usability and subjective user perceptions. While UX evaluation criteria have traditionally been ambiguous, the inclusion of 29 heuristics in Track 1 and the construction of UX metrics in Track 2 offer clear and structured guidance on essential evaluation items, ensuring comprehensive coverage.

Moreover, this UXEM is well-suited for most B2B SaaS product development projects, as it emphasizes universal applicability and customization possibilities when selecting and integrating methods and tools. However, since it is not initially tailored for specific projects, additional efforts—such as applying personas and user

Table 2: Interview Procedure

Category	Question	Content
Background Information	Q1	Briefly describe your professional background and experience in user experience.
	Q2	Please tell me about your specific research or practical experience in user experience evaluation.
Preliminary Understanding	Q3	Have you used similar user experience evaluation methods before?
	Q4	What core elements do you focus on when evaluating user experience?
	Q5	How were your initial impressions about this user experience evaluation method?
Method Evaluation	Q6	What do you think is the potential for application in theory and practice?
	Q7	What are the advantages in the B2B SaaS development?
	Q8	What are the possible limitations or areas for improvement?
	Q9	What challenges do you think might be encountered in practical operations?
In-depth Discussion	Q10	How could this method be better optimized for broader application in UX research?
	Q11	Do you have any specific suggestions for improving this method?

Table 3: Thematic Analysis Results

Theme	Sub-theme	Codes
Advantages and Disadvantages	Advantages (7)	Compatibility Subjective and objective perspectives Comprehensive evaluation Former research based Quantifiable UX standards Feasibility Useful usability categories
	Disadvantage (2)	Not project-specific Extra efforts required
Challenges in Adoption	Challenges in implementation (4)	Complex user roles Restricted user access Absence in practice Evaluator familiarity
	Challenges in team management (8)	Restricted time Designer Leadership Designer's focuses PM's background PM's focuses Staff shortages Acknowledgement within the team High barriers of learning
Suggestions	Suggestions for improvement (5)	Evaluation by other divisions Customization Duplicate effort elimination Integrating project schedules Visualized report

journey maps—are necessary during implementation to address the unique needs of individual projects. Admittedly, while customization can be time-consuming in practical applications, it is critical for effective UX evaluation, as specific usage scenarios and user differences inherently influence assessments. No universal method can function as a one-size-fits-all solution for all evaluation steps without some degree of customization. Although this study does not propose a definitive approach to item customization in Track 1, Track 2 provides UX evaluators with a robust workflow to construct UX metrics.

5.4 Challenges in the adoption of this UXEM

The lack of practical experience and negative perceptions of service design tools may hinder the adoption of this UXEM. Interviewee A criticized these tools as time-consuming and occasionally irrelevant,

particularly for smaller companies or projects operating under tight schedules. In contrast, Interviewee B, actively developing and educating on internal user testing methodologies, acknowledged the benefits of service design tools in UX evaluation. However, B also highlighted the significant challenges in training non-UX colleagues to use these tools effectively, revealing that for teams with limited UX experience, a higher barrier to expertise acquisition may hinder adoption within the team.

Additionally, successfully implementing this UXEM often depends on the leadership of UX designers and the organization’s priorities. For example, a significant challenge arises from the divergent focuses of product managers and UX designers. Interviewees A and B observed that while UX designers prioritize enhancing the UX, product managers often emphasize rapid product release and iteration to achieve favorable conversion rates. This disparity

underscores that the effective adoption of this UXEM is closely tied to the company's structure and commitment to UX. Interviewee A noted that introducing and implementing effective UX strategies becomes particularly difficult in organizations where revenue takes precedence over user-centered design or where UX designers hold a relatively low status.

5.5 Improvement Opportunities

Regarding suggestions for improvement, Interviewee A emphasized the need for a more visualized report format, as it would be preferable to abstract numerical data. A also highlighted the importance of adopting an agile approach to creating personas and user journey maps, given that these tools are often time-consuming. Interviewee B suggested that prioritizing usability issues should account for the project schedule and the functional importance of features, ensuring a balanced approach. Additionally, B recommended streamlining processes to minimize duplicate efforts during iterative development cycles. Interviewee C proposed further customization of Track 1, the usability heuristic evaluation, to better align with the unique requirements of specific B2B SaaS projects.

6 DISCUSSION

Compared to related work, Dual-Track UX expands the application of the GSM process [19] to the B2B domain in a more context-specific manner. While the original GSM process was primarily designed for large-scale UX evaluations of B2C products, focusing on financial metrics such as conversion rates and ROI, Dual-Track UX leverages the GSM process's strengths in enabling long-term UX evaluation and adopts a more user-centered approach by introducing service design tools, which are effective in addressing complex user roles and operational patterns.

Additionally, Dual-Track UX addresses a significant gap in existing UX evaluation frameworks and methods: the lack of consideration for restrictions on external user access. For example, although the FRAMUX-EV framework [14] offers numerous UX-EMs for agile development, it lacks effective integration mechanisms to accommodate scenarios where external user involvement is limited. Dual-Track UX enhances evaluation workflows by incorporating usability heuristic evaluation and exploratory testing into a dual-track structure, thereby minimizing the reliance on full user participation.

Furthermore, this study also establishes a practical approach to UXEM development, which could serve as a valuable reference for researchers working on UXEMs in similar domains.

According to the feedback gathered through interviews, while the broad applicability of this method has been recognized, there remains a need to optimize the coordination of UX evaluations with specific project timelines further, notably by tailoring Dual-Track UX to align with each project's development priorities. Additionally, in the iterative development process of SaaS products, addressing the issue of reducing repetitive tasks during UX evaluations has been identified as an area for improvement. Furthermore, regarding the application of service design tools, some view them as time-consuming in time-restricted projects. Future iterations should focus on adopting rapid techniques for creating personas and user journey maps to enhance efficiency.

Regarding the limitations, the effectiveness of Dual-Track UX in real-world projects remains uncertain, as the validation conducted in this research focused on gathering preliminary feedback through interviews to refine its initial version.

7 CONCLUSION

UX evaluation is critical for identifying user requirements and mitigating the risk of product failure during the development of interactive systems. This paper presents Dual-Track UX, a specialized UX evaluation method designed to address three key challenges in evaluating the UX of B2B SaaS products: (a) diversified user goals and operational patterns, (b) limited access to external users, and (c) the necessity for long-term evaluation. Specifically, Dual-Track UX integrates heuristic evaluation, GSM process, exploratory testing, and service design tools into a cohesive dual-track workflow. Preliminary validation through expert interviews highlighted the method's advantages, limitations, challenges in adoption, and opportunities for further refinement, paving the way for its broader application in B2B SaaS development. The authors plan to refine Dual-Track UX into version 2.0 for the next steps, incorporating feedback obtained from expert interviews. Following this iteration, the updated method will be applied to a real-world B2B SaaS product development project in collaboration with industry partners to evaluate its effectiveness and practical applicability further.

Acknowledgments

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